

Erdős Problem #138

EP-138

L1 - TRACTABLE

OPEN

Number Theory

ORIGINAL PROBLEM STATEMENT

Let the van der Waerden number $W(k)$ be such that whenever $N \geq W(k)$ and $\{1, \dots, N\}$ is 2-coloured there must exist a monochromatic k -term arithmetic progression. Improve the bounds for $W(k)$ - for example, prove that $W(k)^{1/k} \rightarrow \infty$.

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>